TED UNIVERSITY

Faculty of Engineering

Department of Computer Engineering

CMPE 491 Senior Project

ARIS Project Proposal

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ARIS

Project Web Page:

https://ezeynepel.github.io/Senior-Project/

Project Goal

The aim of this project is to develop an artificial intelligence supported smart helmet radio that will interpret the facial expressions, gestures and voice commands of soldiers in order to provide real-time communication, coordination and security in military operations.

Project Idea Description

ARIS is an advanced smart helmet system designed to provide field communication for soldiers. The system converts gesture and facial expression recognition with voice command processing into a wearable communication module by mounting it on military helmets.

The helmet contains a miniature camera that continuously captures facial expressions and hand movements. The system transforms these inputs into relevant commands through machine learning-based AI image processing. A speech recognition module constantly listens for pre-programmed voice commands and provides instantaneous interpretation. Soldiers can communicate without the use of manual equipment or physical contact because of ARIS's integration of these two systems, which facilitates effective and continuous multimodal interaction.

All the data that's processed is securely transmitted via encrypted connections to either a central command system or other units in the field. Even under tough conditions, the communication system is designed to deliver reliable, low-latency performance.

During the implementation phase of the project, hardware components such as an embedded system consisting of a microcontroller, camera, microphone, and wireless communication modules for multi-modal signal interpretation will be integrated with Al-driven software modules. Design will be impacted with a modular software architecture that can be easily configured and scaled in the future (e.g., environmental awareness or sentiment analysis).

ARIS will evolve through iterative design and test phases until the final objective of completing a functional prototype for real-time multi-modal command recognition can be achieved. The prototype should achieve maximum communication velocity, security, and coordination efficiency for the people in the field.